

Building Systems With Pixels



Dilin Anand is a senior assistant editor at EFW

All the education in this world could not stop me from burning out parts while working on an electronics project. But being able to reduce the chances of having expensive components go up in smoke would be welcome any day. The option to virtually design electronics has this as one of its many benefits.

We already have exceptionally well-designed software designed to help beginners get started with electronics virtually, without having to directly handle the tools or equipment themselves.

Virtual electronics does not stop there though. It can help you through a number of things from being a platform for collaboration between research and development teams sitting on opposite sides of the planet, to being able to let someone test your chip as part of his or her system without having to give complete access to the component itself [intellectual property (IP) protection].

Creating virtual situations

At Mobile World Congress (MWC) 2015 in Barcelona, two simulators were used to set

up a test system that allowed engineers to simulate any global location and mobile connectivity on demand.

Smart Systems Laboratory of University of Hertfordshire, the UK, wanted to demonstrate a connected car and its diagnostics systems in action at MWC. They did this by using a satellite-positioning technology called global navigation satellite system (GNSS) through a simulator from Spectracom. This was combined with simulated long term evolution (LTE) network environment provided by Anritsu's mobile network simulator. The connected car system being developed is based on the wireless transmission of onboard diagnostics (OBD) information available through a cloud server using mobile backhaul (LTE, 3G and GPRS) together with global positioning system (GPS) location.

Rahman Jamal, global technology and marketing director, National Instruments (NI), explains that, "As things get complex, say, in a car, for example, you want to do as much engineering as possible in the laboratory where it is safe to test.

This is where hardware-in-the-loop (HIL) testing is done. Once that piece of software is running in the loop, it acts as if it was already built. If this test is working out, you can start replacing parts of the simulation with real-built hardware."

This is something that was known for the automotive industry but it is now being used for white goods like refrigerators and washing machines. We have heard from the industry that integration of simulation capabilities with test is something that they need big time. "While circuit design simulation is a very specific area, this is also done while embedding mathematical models of refrigerators and engine control units (ECUs)," adds Jamal.

How simulation helps

Time-to-market is one of the most crucial factors for new product development. An example here is addressing how system-level EDAs can help in this aspect. One of the most popular switched mode power supply topology is called flyback converter. A flyback converter design typically takes an engineer from a few days to a few weeks. "A tool like PSW allows a designer to complete a nearly production-ready design, that is, schematics, bills of materials, magnetic build sheets, plus efficiency/loss analyses, stability and transient response simulations in mere minutes."

—Jason Guo, senior technical staff member, Fairchild Semiconductor

Leveraging advances in consumer electronics

"Over the years, tools like MultiSim have been used to design circuits and do simulation. It even helps people do co-simulation where you can have real circuitry and see how it is different from what you had simulated and then correlate those designs. Taking that to the next level and leveraging newer-generation touch-enabled devices coming in, we have moved part of the MultiSim platform to the iPad. So, people today do not have to wait to have access to their workstations to do simulations."

—Satish Mohanram, technical marketing manager, National Instruments